

Jeremy Williams

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Summary

Staff Data Engineer with 10+ years of experience designing, developing, and implementing scalable streaming and batch ETL/ELT pipelines. Proven track record of architecting large-scale big data systems, driving enterprise FinOps, and mentoring engineering teams. Adept at bridging the gap between complex technical requirements and high-impact business outcomes, with a focus on operational efficiency, data governance, and automated pipeline generation.

Skills and Technology

- **Data Processing & Streaming:** Apache Kafka, Apache Spark, Apache Airflow, MirrorMaker
- **Databricks Ecosystem:** Databricks Unity Catalog, Delta Tables, Apache Spark (Databricks)
- **Programming Languages:** Scala, SQL, Python, Java
- **Cloud & Infrastructure:** Amazon Web Services (S3, EC2, Lambda, SNS, SQS), Linux (Ubuntu, Redhat)
- **Databases:** Elasticsearch, Hive, InfluxDB, MySQL, MS SQL
- **Development & CI/CD:** GitHub, Jenkins, VS Code, Claude Code (AI-Assisted Dev)
- **Project Management:** Jira, Confluence, Mural

Experience

Staff Data Engineer – Visa Inc. *Mar 2020 – Present*

- Mentor a team of 6 data engineers in advanced development practices, spearheading an agentic coding initiative that integrated Jira with Claude Code to autonomously resolve tickets and generate GitHub PRs.
- Drive the architectural vision for Big Data systems, developing and implementing scalable, flexible, and future-proof data processing pipelines across the enterprise.
- Organize and lead 40+ small-group technology training sessions for Kibana and Databricks, democratizing data access.
- Partner with product and engineering leadership to define technical roadmaps, translating complex business visions into actionable architecture.

Key Initiatives & Enterprise Impact:

- **Enterprise Hub-and-Spoke Architecture:** Architected and deployed a massive-scale (10 billion daily records / 2TB) Kafka and MirrorMaker ecosystem to consolidate decentralized data streams, establishing a centralized platform for hundreds of cross-functional users (Product, Security, Ops).
- **Post-Acquisition Datacenter Integration:** Directed the data architecture strategy for migrating and co-locating legacy services into Visa's primary on-premise datacenters, ensuring seamless data interoperability during a massive infrastructure transition.
- **Enterprise FinOps & Cloud Cost Optimization:** Spearheaded a cost-reduction initiative that decreased AWS and Databricks spend by 35% (saving \$1.5M annually) by purging unused datasets, right-sizing Spark clusters, and implementing S3 storage tiering. Further optimized platform economics by architecting a highly reliable single-region infrastructure that exceeds Tier 2 (4-hour RTO) SLAs without the overhead of multi-region redundancy.

- **Global Data Localization (KSA):** Owned the technical execution of a high-stakes data localization mandate for the Kingdom of Saudi Arabia, designing an isolated regional spoke cluster that ensured strict regulatory compliance while integrating localized metrics into the global hub.
- **AI-Driven Pipeline Automation:** Accelerated the rollout of international datacenter pipelines by integrating AI-assisted development (Claude Code) to automatically generate and deploy bronze-level processing code from standardized templates.

Sr. Data Engineer / Data Engineer – CardinalCommerce, A Visa Solution *Jan 2016 – Mar 2020*

- Architected and developed a Kafka and Spark application log streaming pipeline from proof-of-concept to production in under 3 months.
- Planned and deployed a multi-node Apache Airflow cluster, driving a 300% improvement in parallel task execution.
- Engineered and optimized Spark batch and streaming data processing pipelines using Scala and SQL.
- Maintained strict documentation for naming conventions, schema enforcement, deployment, and data pipeline flows.
- Administered and scaled Airflow, Elasticsearch, InfluxDB, and Kafka clusters.

Key Initiatives & Enterprise Impact:

- **Schema-Enforced Streaming:** Pioneered an early iteration of a schema-enforced hub-and-spoke data streaming model using Kafka, Spark, and custom connectors, drastically improving system flexibility and accuracy while reducing mean time to display.
- **Unified Observability:** Centralized application and server metric data for unified dashboarding and alerting using Telegraf, InfluxDB, Kapacitor, and Grafana, giving engineers a single pane of glass view of system and application health.

Awards

- **Recipient:** 2022 CardinalCommerce Technology and Innovation award

Education

- **B.Sc. Physics:** Kent State University, Kent, OH – May 2009